

Problem 8.22

Show that $d_n \leq i_n$.

Solution

By definition, the effective rate of interest in the n th period is

$$i_n = \frac{a(n) - a(n-1)}{a(n-1)},$$

and the effective rate of discount in the n th period is

$$d_n = \frac{a(n) - a(n-1)}{a(n)}.$$

Since the accumulation function is nondecreasing, we know that

$$a(n-1) \leq a(n).$$

Therefore,

$$\frac{1}{a(n)} \leq \frac{1}{a(n-1)}.$$

Also, the amount of growth during the n th period is nonnegative:

$$a(n) - a(n-1) \geq 0.$$

Multiplying both sides of

$$\frac{1}{a(n)} \leq \frac{1}{a(n-1)}$$

by the nonnegative quantity $a(n) - a(n-1)$ gives

$$\frac{a(n) - a(n-1)}{a(n)} \leq \frac{a(n) - a(n-1)}{a(n-1)}.$$

Thus,

$$\boxed{d_n \leq i_n}.$$